

Data Mining (EECS 6412)

Introduction to Data Mining
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Outline of Introduction

- ▶ Why data mining?
- ▶ What is data mining?
- ▶ Process of data mining or KDD
- ▶ Course information
- ▶ Overview of data mining
 - ▶ What kind of data to mine from?
 - ▶ What kind of patterns to mine? (Data Mining Tasks)
 - ▶ Data mining R&D issues

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Why Data Mining?

- ▶ Data mining started around 1990
- ▶ People were from
 - ▶ Database systems, machine learning (ML), and statistics
- ▶ ML at that time
 - ▶ Worked only on small datasets
 - ▶ Focused on classification, regression and clustering tasks
- ▶ There was need to
 - ▶ Handle large amounts of data
 - ▶ Learn other types of useful patterns

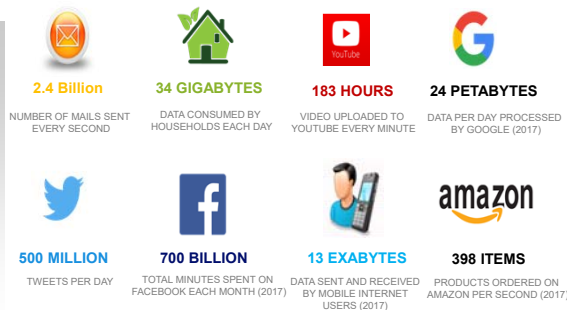
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Why Data Mining Became Popular?

- ▶ Data flood
 - ▶ Vast amounts of data are produced every day
- ▶ Data are generated by:
 - ▶ Bank, telecom, stores, other business transactions ...
 - ▶ Scientific fields: astronomy, biology, etc
 - ▶ Web and social media (text, images, audio, video, etc.)
 - ▶ Internet of Things (IoT)
 - ▶ In 2020, 31 billion IoT devices exist (source: Statista).
- ▶ Big Data Era

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Big Data Examples



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Characteristics of Big Data

- ▶ 5Vs in Big Data:
 - ▶ Volume
 - ▶ Velocity
 - ▶ Variety
 - ▶ Veracity
 - ▶ Value

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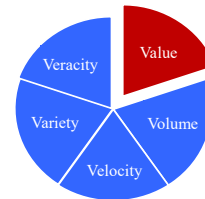
Value

- ▶ Valuable information hidden in data
- ▶ For example:
 - ▶ What types of credit card transactions are fraud?
 - ▶ What items are often bought together?
 - ▶ What types of customers like a particular product?
 - ▶ What news or ads should be recommended to an online news reader?
 - ▶ Who are the most influential people in a social network?
 - ▶ What topics are people discussing in social media?
 - ▶ What types of patients react positively to a new drug?

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Purpose of Data Mining

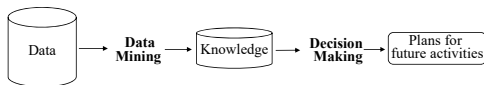
- ▶ Data mining is a tool to get the value out of data.



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What is Data Mining?

- ▶ Mining knowledge from data
 - ▶ Data = raw information
 - ▶ Knowledge = patterns or models 'behind' the data



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What Is Data Mining?

- ▶ Data mining [Han, 2001]
 - ▶ process of extracting interesting (*non-trivial, implicit, previously unknown* and *potentially useful*) knowledge or patterns from data in *large* databases.
- ▶ Objectives of data mining:
 - ▶ Discover knowledge that characterizes general properties of data
 - ▶ Discover patterns on the previous data in order to make predictions on future data

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Example 1

- ▶ Data about investors

Customer ID	Account Type	Margin Account	Transaction Method	Trades/Month	Sex	Age	Favorite Recreation	Annual Income
1005	Joint	No	Online	12.5	F	30-39	Tennis	40-59K
1013	Custodial	No	Broker	0.5	F	50-59	Skiing	80-99K
1245	Joint	No	Online	3.6	M	20-29	Golf	20-39K
2110	Individual	Yes	Broker	22.3	M	40-49	Fishing	40-59K
1001	Individual	Yes	Online	5.0	M	30-39	Golf	60-79K

- ▶ Possible business questions
 - ▶ Can we develop a general characterization/profile of different investor types? (characterization/frequent patterns)
 - ▶ What characteristics distinguish between Online and Broker investors? (classification)
 - ▶ Can we develop a model which will predict the average trades/month for a new investor? (regression)

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Example 2

Hypothetical Training Data for Disease Diagnosis

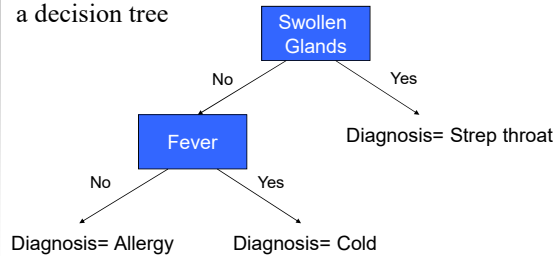
Patient ID#	Sore Throat	Fever	Swollen Glands	Congestion	Headache	Diagnosis
1	Yes	Yes	Yes	Yes	Yes	Strep throat
2	No	No	No	Yes	Yes	Allergy
3	Yes	Yes	No	Yes	No	Cold
4	Yes	No	Yes	No	No	Strep throat
5	No	Yes	No	Yes	No	Cold
6	No	No	No	Yes	No	Allergy
7	No	No	Yes	No	No	Strep throat
8	Yes	No	No	Yes	Yes	Allergy
9	No	Yes	No	Yes	Yes	Cold
10	Yes	Yes	No	Yes	Yes	Cold

- In this example dataset, attributes correspond to Symptoms, and Diagnosis.
- The natural question is to predict the Diagnosis (class) [the Output variable] from the symptoms [the input variables].

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Example 2 (Cont'd)

Pattern discovered:
a decision tree



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Example 2 (Cont'd)

Using Decision Tree for Prediction

Data Instances with an Unknown Classification

Patient ID#	Sore Throat	Fever	Swollen Glands	Congestion	Headache	Diagnosis
11	No	No	Yes	Yes	Yes	?
12	Yes	Yes	No	No	Yes	?
13	No	No	No	No	Yes	?

What are the predicted diagnoses?

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Example 3

Part of training data for Iris plants classification



Sepal Length	Sepal Width	Petal Length	Petal Width	Class
5.1	3.5	1.4	0.2	Iris Setosa
4.9	3	1.4	0.2	Iris Setosa
4.7	3.2	1.3	0.2	Iris Setosa
5.4	3.9	1.7	0.4	Iris Setosa
4.8	3	1.4	0.1	Iris Setosa
7	3.2	4.7	1.4	Iris Versicolour
6.4	3.2	4.5	1.5	Iris Versicolour
5.5	2.3	4	1.3	Iris Versicolour
5	2	3.5	1	Iris Versicolour
5.4	3	4.5	1.5	Iris Versicolour
6	3	4.8	1.8	Iris Virginica
6.9	3.1	5.4	2.1	Iris Virginica
6.5	3	5.2	2	Iris Virginica
6.2	3.4	5.4	2.3	Iris Virginica
5.9	3	5.1	1.8	Iris Virginica

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Generated Rules for Iris Plants Classification

Rules for Class = Iris Setosa (1 rule)

Rule 1: (petal_length <= 1.9) → (class = Iris Setosa)

Rules for Class = Iris Versicolour (3 rules)

Rule 1: (1.9 < petal_length <= 4.9) ∧ (petal_width <= 1.6) → (class = Iris Versicolour)

Rule 2: (sepal_length > 5.8) ∧ (petal_length <= 5.1) ∧ (petal_width <= 1.7) → (class = Iris Versicolour)

Rule 3: (sepal_length > 5.8) ∧ (sepal_width > 3.0) ∧ (petal_length <= 4.8) → (class = Iris Versicolour)

Rules for Class = Iris Virginica (4 rules)

Rule 1: (petal_length > 4.8) ∧ (petal_width > 1.7) → (class = Iris Virginica)

Rule 1: (sepal_length > 6.2) ∧ (petal_length > 5.0) → (class = Iris Virginica)

Summary: Total number of rules = 8

Average length of rules = 2.75

Training accuracy = 99.33%

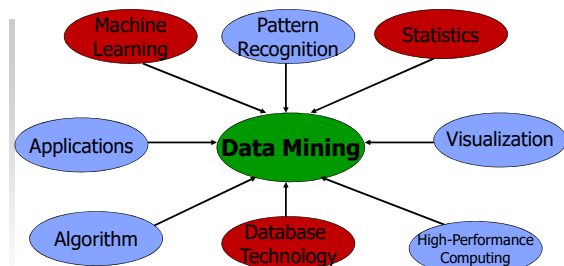
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Example 4: Recommendation

- ▶ A person buys a book (product) at amazon.com.
- ▶ Task:
 - ▶ Recommend other books (products) this person is likely to buy
- ▶ One method:
 - ▶ Use frequent pattern mining on transaction histories and may find:
 - ▶ Customers who bought "Advances in Knowledge Discovery and Data Mining", also bought "Data Mining: Concepts and Techniques"
- ▶ Recommendation program is quite successful

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Data Mining: Confluence of Multiple Disciplines



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Some Related Fields

- ▶ Machine Learning
 - ▶ Subfield of AI
- ▶ Exploratory data analysis
 - ▶ Subfield of Statistics
- ▶ Data warehousing and OLAP (Database)
 - ▶ Data summarization and aggregation
- ▶ Data mining (in comparison to ML)
 - ▶ inherits techniques from ML and statistics
 - ▶ emphasizes processing very large data sets
 - ▶ finds more types of patterns

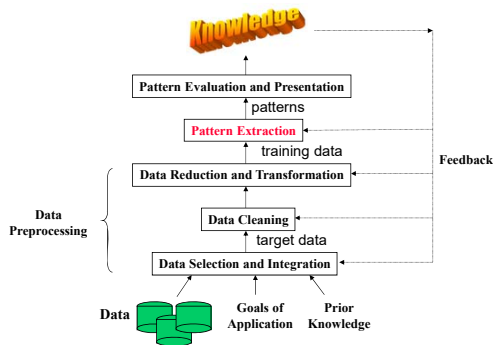
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Alternative Name: KDD

- ▶ Knowledge discovery in databases (KDD)
 - ▶ Used by AI, Machine Learning Community since 1989
- ▶ Data mining
 - ▶ Used by DB, business people since 1990
- ▶ Two names are now used interchangeably

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Process of Data Mining and KDD



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Course Objectives

- ▶ To provide an introduction to data mining
 - ▶ To understand the fundamental concepts, principles, algorithms, and applications of data mining.
- ▶ To gain hands-on experience by
 - ▶ implementing some data mining algorithms.
 - ▶ using data mining software to mine patterns from real world data
- ▶ To study advanced topics in data mining
- ▶ To prepare yourself for
 - ▶ pursuing research in data mining,
 - ▶ using data mining in your research or pursuing data analytics jobs

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Course Content

- ▶ Basics of data mining
 - ▶ Pattern mining
 - ▶ Frequent pattern and association rule mining
 - ▶ Sequential pattern mining
 - ▶ Mining classification models
 - ▶ Data preprocessing (feature selection, normalization, missing value handling, etc.)
 - ▶ Decision tree learning
 - ▶ Bayesian classification (naïve Bayes and Bayesian networks)
 - ▶ Neural networks and introduction to deep learning
 - ▶ K-nearest neighbor method
 - ▶ Learning from imbalanced data
 - ▶ Clustering analysis
- ▶ Text classification
- ▶ Graph mining
- ▶ Advanced topics (e.g., recommendation systems, multi-modal sentiment analysis, knowledge graphs, causality discovery, etc.)

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Reference Books and Materials

- ▶ Reference books
 - ▶ Jiawei Han, Jian Pei and Hanghang Tong, *Data Mining: Concepts and Techniques*, Morgan Kaufmann, 4th Edition, 2023.
 - ▶ Pang-Ning Tan, Michael Steinbach, Anuj Karpatne, Vipin Kumar, *Introduction to Data Mining*, Second Edition, Pearson, 2019.
 - ▶ Charu C. Aggarwal, *Data Mining, The Textbook*, Springer, 2015.
 - ▶ Ian H. Witten and Eibe Frank, *Data Mining: Practical Machine Learning Tools and Techniques (Second Edition)*, Morgan Kaufmann, 2005.
- ▶ Some conference/journal papers
 - ▶ (To be) posted on the course web site

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Evaluation and Grading

- ▶ Assignments (20%)
 - ▶ 2 assignments
- ▶ Project (45%)
- ▶ Final exam (25%)
- ▶ Participation (10%)

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Project

- ▶ Should be done in a group of 2 students.
- ▶ Pick a recent advanced topic in data mining
 - ▶ Topic list will be released in the second week
- ▶ Conduct a survey on the topic
 - ▶ Present the survey to the class in November (10%)
- ▶ Implement and evaluate a (SOTA) method as a baseline
 - ▶ Submit code and a short report (10%)
- ▶ Design your solution to the problem
 - ▶ E.g., a reasonable improvement to the baseline
- ▶ Implement your solution
- ▶ Evaluate your solution by comparing it to baseline(s)
- ▶ Write a final project report (in a research paper format) (20%)
- ▶ Present the project to the class (7-10 minutes) (5%)

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Survey Presentation

- ▶ Make a 30-minute presentation to the class
 - ▶ What is the topic area about?
 - ▶ Motivation, objectives, and problem definition
 - ▶ What are the categories of existing techniques?
 - ▶ Briefly present a representative method in at least two categories.
 - ▶ What are the advantages of each category or technique?
 - ▶ What are the limitations of each category or technique?
 - ▶ What are the open (unresolved) issues in the research?
- ▶ Purpose:
 - ▶ Make the class understand the topic areas and techniques for solving the problem
 - ▶ Develop presentation skills

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Instructor Information

- ▶ Office hours
 - ▶ Thursdays: 1:00pm-2:00pm
 - ▶ Location: LAS 2048
- ▶ Email: aan@yorku.ca
- ▶ Course eClass site:
 - ▶ <https://eclass.yorku.ca/course/view.php?id=144958>

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What Kind of Data?

- ▶ Tabular data
- ▶ Transactional data
- ▶ Text data
- ▶ Time-series data
- ▶ Sequence data
- ▶ Data streams
- ▶ Graphs
- ▶ Multimedia databases
- ▶ ...

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Tabular Data

Structured data

- Table
- Examples
- Attributes
- Can be stored in
 - plain text files, or
 - relational databases.
- Most common form of data for classification, clustering and regression tasks

Attributes/Features/Variables

OUTLOOK	TEMPERATURE	HUMIDITY	WINDY	PLAY
sunny	85	85	FALSE	Don't Play
sunny	80	90	TRUE	Don't Play
overcast	83	78	FALSE	Play
rain	70	96	FALSE	Play
rain	68	80	FALSE	Play
rain	65	70	TRUE	Don't Play
overcast	84	65	TRUE	Play
sunny	72	95	FALSE	Don't Play
sunny	69	75	FALSE	Play
rain	75	80	FALSE	Play
sunny	75	70	TRUE	Play
overcast	72	90	TRUE	Play
overcast	81	75	FALSE	Play
rain	71	80	TRUE	Don't Play

An attribute can be:

- Symbolic (discrete, categorical)
- Numeric (continuous, integer)
- Ordinal (categorical with a value order, e.g., temperature with values of low, medium, high)

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Transactional Data

Transaction-id	Itemset
T100	Milk, bread, beer, diaper
T200	Beer, cook, fish, potato, orange, diaper
...	...

Some transactional databases also contain time stamp and the customer id for each transaction

Patterns

- What kind of product combinations that customers like to buy together?

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Text Data

- Documents
 - articles, Web pages, blogs, tweets, emails, product specifications, reports, notes, reviews, etc.
- Structure
 - highly unstructured (news, stories, etc.), or
 - semistructured (HTML/XML documents, etc.)
- What can be discovered from a text database?
 - text classification models
 - keyword or content associations
 - summaries, topics, sentiments, emotions
 - ...

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Example of XML Data

```
<breakfast_menu>
  <food>
    <name>Belgian Waffles</name>
    <price>$5.95</price>
    <description>Two of our famous Belgian Waffles with plenty of real maple syrup</description>
    <calories>650</calories>
  </food>
  <food>
    <name>Strawberry Belgian Waffles</name>
    <price>$7.95</price>
    <description>Light Belgian waffles covered with strawberries and whipped cream</description>
    <calories>900</calories>
  </food>
  <food>
    <name>Berry-Berry Belgian Waffles</name>
    <price>$8.95</price>
    <description>Light Belgian waffles covered with an assortment of fresh berries and whipped cream</description>
    <calories>900</calories>
  </food>
  <food>
    <name>French Toast</name>
    <price>$4.50</price>
    <description>Thick slices made from our homemade sourdough bread</description>
    <calories>600</calories>
  </food>
  <food>
    <name>Homestyle Breakfast</name>
    <price>$6.95</price>
    <description>Two eggs, bacon or sausage, toast, and our ever-popular hash browns</description>
    <calories>950</calories>
  </food>
</breakfast_menu>
```

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Time Series Data

- A sequence of values that change with time
 - Daily water consumption data in a city
 - Data collected regarding the stock exchange
- Types of analysis
 - Trend analysis
 - To predict future values
 - Similarity search
 - Find similarities in sub-sequences, such as periodic patterns, recurring patterns

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Sequence Data

- Sequences of ordered objects or events (with or without concrete notation of time)
 - Bio-sequences
 - DNA, protein
 - Web log data
 - Click stream (web page traversal sequences)
 - Sequences of items bought by a customer
- Patterns
 - Frequent sequences, alignments of two bio-sequences, etc.

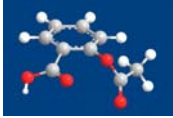
upstream → downstream
 ... CCTTACTTATAATCTCATCTA ...
 ... GGAAATGAATATTACGATCATAT ...
 downstream ← upstream
 An example DNA sequence and its paired strand
 (Credit: Gerardo M Nava)

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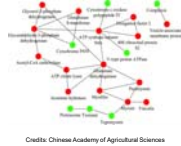
Graph Data

- ▶ Data entities are connected, forming graphs

Aspirin



Biological Networks

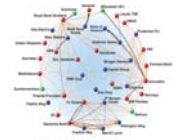


Credit: Chinese Academy of Agricultural Sciences

Social Networks



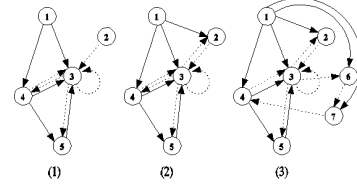
Financial Networks



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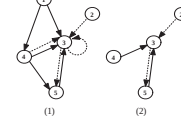
Example 1 of Graph Dataset

Graph data set: containing one or more graphs (3 in this example)



Patterns can be found:

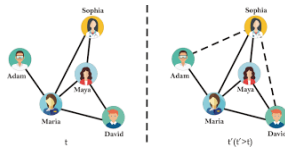
- Frequent subgraphs



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Example 2 of Graph Dataset

Social networks



Two graph mining tasks:

- Link prediction
- Node classification

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Other Graph Mining Tasks

- ▶ Find communities in social networks
- ▶ Find the most influential people in communities
- ▶ Anomaly detection
- ▶ Etc.

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Basic Data Mining Tasks

- ▶ **Predictive:** *Discover patterns on previous & current data in order to make predictions on future data*
 - ▶ Classification
 - ▶ Regression
 - ▶ Structured prediction
- ▶ **Descriptive:** *Discover knowledge that characterizes general properties of data*
 - ▶ Clustering
 - ▶ Concept characterization / Summarization
 - ▶ Association analysis (frequent itemsets, association rules)
 - ▶ Sequential pattern mining
- ▶ **Predictive or descriptive**
 - ▶ Time series analysis
 - ▶ Outlier detection
 - ▶

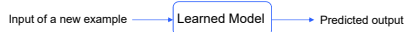
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Predictive Tasks

- ▶ A form of **Supervised Learning**
 - Training examples are **labeled** with target outputs

Patient ID	Input/Independent/Condition attributes/features/variables					Target/Output/Decision attribute/variable
	Sore Throat	Fever	Swollen Glands	Congestion	Headache	Diagnosis
1	Yes	Yes	Yes	Yes	Yes	Strep throat
2	No	No	No	Yes	Yes	Allergy
3	Yes	Yes	No	Yes	No	Cold
4	Yes	No	Yes	No	No	Strep throat
5	No	Yes	No	Yes	No	Cold
6	No	No	No	Yes	No	Allergy
7	No	No	Yes	No	No	Strep throat
8	Yes	No	No	Yes	Yes	Allergy
9	No	Yes	No	Yes	Yes	Cold
10	Yes	Yes	No	Yes	Yes	Cold

- Learn a model from labeled data to make predictions on the future data



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Types of Predictive Tasks / Supervised Learning

- ▶ Classification
- ▶ Regression
- ▶ Structured prediction

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Classification

- ▶ Target variable is discrete
- ▶ Its values are considered as *classes (labels)*
 - The target attribute is also called *class attribute*
 - Examples:
 - Diagnosis: {strep throat, allergy, cold}
 - Image type: {cat, dog, car, truck,}
- ▶ Goal:
 - Learn a **model** from the **labeled** training data
 - Learned model is used to classify new examples

Patient ID						Class attribute
	Sore Throat	Fever	Swollen Glands	Congestion	Headache	Diagnosis
1	Yes	Yes	Yes	Yes	Yes	Strep throat
2	No	No	No	Yes	Yes	Allergy
3	Yes	Yes	No	Yes	No	Cold
4	Yes	No	Yes	No	No	Strep throat
5	No	Yes	No	Yes	No	Cold
6	No	No	No	Yes	No	Allergy
7	No	No	Yes	No	No	Strep throat
8	Yes	No	No	Yes	Yes	Allergy
9	No	Yes	No	Yes	Yes	Cold
10	Yes	Yes	No	Yes	Yes	Cold

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Classification (Cont'd)

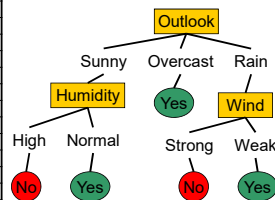
- ▶ Models can be in different types
 - Decision tree
 - A set of rules
 - Bayesian network
 - Support vector machine
 - Neural network
 -
- ▶ For different types of models, different types of learning algorithms were designed

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Example of Classification Models

- ▶ A decision tree for *PlayTennis*

Day	Outlook	Temp	Humid	Wind	PlayTennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No



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Types of Predictive Tasks / Supervised Learning

- ▶ Classification
- ▶ Regression
- ▶ Structured prediction

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Regression

► Target variable is numeric, e.g.,

- House price
- Daily water consumption in a city
- Exam score

► Goal:

- Learn a model from the *labeled* training data to model the relationship between the input and target variables
- Learned model is used to predict a numeric value for a new example

Input/independent variables/predictors

Target attribute/variable (Dependent variable)

Hours	Prep exams	Exam score
1	1	76
2	3	78
2	3	85
4	5	88
2	2	72
1	2	69
5	1	94
4	1	94
2	0	88
4	3	92
4	4	90
3	3	75
6	2	96
5	4	90
3	4	82
4	4	85
6	5	99
2	1	83
1	0	62
2	1	76

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Regression

► Linear vs. non-linear regression

- Relationship between independent and dependent variables are linear or non-linear

► Simple vs. multiple regression

- Simple regression
 - only one independent variable
- Multiple regression
 - multiple independent variables

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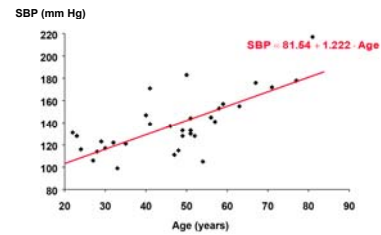
Simple Linear Regression - Example

Data: Age and systolic blood pressure (SBP) among 33 adult women

Age	SBP	Age	SBP	Age	SBP
22	131	41	139	52	128
23	128	41	171	54	105
24	116	46	137	56	145
27	106	47	111	57	141
28	114	48	115	58	153
29	123	49	133	59	157
30	117	49	128	63	156
32	122	50	163	67	176
33	99	51	130	71	172
35	121	51	133	77	178
40	147	51	144	81	217

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Simple Linear Regression - Example



adapted from Colton T. Statistics in Medicine. Boston: Little Brown, 1974

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Multiple Linear Regression Example

Target attribute/variable

Hours	Prep exams	Exam score
1	1	76
2	3	78
2	3	85
4	5	88
2	2	72
1	2	69
5	1	94
4	1	94
2	0	88
4	3	92
4	4	90
3	3	75
6	2	96
5	4	90
3	4	82
4	4	85
6	5	99
2	1	83
1	0	62
2	1	76

► Linear regression model learned:

$$\text{Exam score} = 67.67 + 5.56 \cdot (\text{Hours}) - 0.60 \cdot (\text{Prep exams})$$

► Prediction:

- A new student with Hours=4 and Prep exams = 2
- Predicted exam score:

$$\text{Exam score} = 67.67 + 5.56 \cdot 4 - 0.60 \cdot 2 = 88.71$$

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Types of Predictive Tasks / Supervised Learning

- Classification
- Regression
- Structured prediction

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Structured Prediction

- ▶ Outputs are structured objects, e.g.,
 - Sequences, trees, graphs, or other interrelated components
 - There are often relationships among output components
- ▶ Example:
 - Sequence-to-sequence problems
 - Both input and output are a sequence
 - E.g., machine translation (English → French)

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Structured Prediction

- Another example: Named entity recognition

Labeled Training data:

John **PERSON** joined **COMPANY** as a data scientist in **Milton Keynes**, after finishing his computer science degree at the **University of Lincoln**.

John works in the **IT** department, which is the internal software development team at **Google** office in **Milton Keynes**. It focuses on developing a next-generation software product called **InsightMaker**, which uses AI to manage information intuitively and discover valuable insights.

Prediction on new data:

When **Sebastian Thrun** **PERSON** started working on self-driving cars at **Google** **COMPANY** in **2007** **DATE**, few people outside of the company took him seriously.

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Basic Data Mining Tasks

- ▶ Predictive:
 - ▶ Classification
 - ▶ Regression
 - ▶ Structured prediction
- ▶ **Descriptive:**
 - ▶ Clustering
 - ▶ Concept characterization / Summarization
 - ▶ Association analysis (frequent itemsets, association rules)
 - ▶ Sequential pattern mining
- ▶ Predictive or descriptive
 - ▶ Time series analysis
 - ▶ Outlier detection
 - ▶

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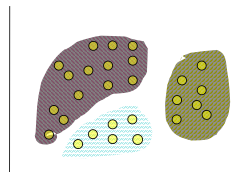
Descriptive Data Mining Tasks

- ▶ Training data do not have output or target variables
- ▶ Goal is to
 - Extract hidden patterns or relationships in data
 - Discover knowledge that characterizes general properties of data
- ▶ A form of **unsupervised learning**
- ▶ Typical descriptive tasks
 - Clustering
 - Concept characterization / Summarization
 - Association analysis (frequent itemsets, association rules)
 - Sequential pattern mining

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Clustering

- ▶ Class label is unknown in the training data.
- ▶ Group data to form classes (clusters)
 - ▶ Unsupervised learning.
- ▶ Principle:
 - ▶ maximizing the intra-class similarity
 - ▶ minimizing the inter-class similarity
- ▶ Applications
 - ▶ Market/customer segmentation
 - ▶ Image segmentation
 - ▶



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Concept Characterization

- ▶ Summarization of general properties of objects in a target group
- ▶ Example:
 - ▶ Characterize customers who spend more than \$1000 a year in the AllElectronic store and have excellent credit ratings.
 - ▶ Result: 40-50 years old, employed,

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Association Analysis

- ▶ Examples of association rules:
 - ▶ $\text{age}(X, "20..29") \wedge \text{income}(X, "20..29K") \rightarrow \text{buys}(X, "PC")$
[support = 2%, confidence = 60%]
 - ▶ $\text{contains}(T, "computer") \rightarrow \text{contains}(T, "software")$
[support=1%, confidence=75%] (T stands for a transaction)
- ▶ Widely used for market basket or transactional data analysis.
 - ▶ What products were often purchased together?

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Mining Sequential Patterns

- ▶ Find frequently occurring patterns in a sequence database.
 - ▶ Within 3 months, buy computer $\rightarrow$ buy USB drive $\rightarrow$ buy digital camera
- ▶ Applications
 - ▶ Sale campaign analysis
 - ▶ What are the subsequent purchases after buying a PC?
 - ▶ Web log analysis
 - ▶ DNA or protein analysis

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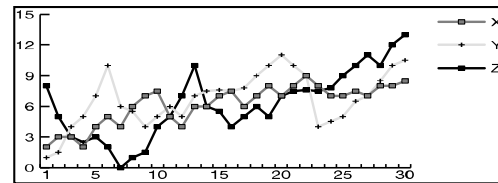
Basic Data Mining Tasks

- ▶ Predictive:
 - ▶ Classification
 - ▶ Regression
- ▶ Descriptive:
 - ▶ Clustering
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 - ▶ Time series analysis
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Time Series Analysis

- ▶ Mine from time-series data to
 - ▶ Predict future values (Predictive)
 - ▶ Determine similar patterns over time (Descriptive)
- ▶ Example: Stock Market



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Outlier/Anomaly Detection

- ▶ Outlier
 - ▶ A data object that does not comply with the general behavior of the data
- ▶ Noise or exception?
 - ▶ One person's garbage could be another person's treasure
- ▶ Methods:
 - ▶ Clustering
 - ▶ Classification
 - ▶ Regression analysis
 - ▶ ...
- ▶ Useful in fraud detection, rare events analysis

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What will be covered in lectures

- ▶ Predictive:
 - ▶ Classification ✓
 - ▶ Regression (will cover regression trees ✓)
 - ▶ Structured prediction
- ▶ Descriptive:
 - ▶ Clustering ✓
 - ▶ Concept characterization / Summarization
 - ▶ Association analysis (frequent itemsets, association rules) ✓
 - ▶ Sequential pattern mining ✓
- ▶ Predictive or descriptive
 - ▶ Time series analysis
 - ▶ Outlier detection

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Questions

- ▶ What is the difference between classification and clustering?
- ▶ What is the difference between classification and regression?

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Outline of Introduction

- ▶ Why data mining?
- ▶ What is data mining?
- ▶ Process of data mining or KDD
- ▶ Course information
- ▶ Overview of data mining
 - ▶ What kind of data to mine from?
 - ▶ What kind of patterns to mine? (Data Mining Tasks)
- ▶ *Data mining R&D issues*

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Major Issues in Data Mining

- ▶ Efficiency
- ▶ Effectiveness
- ▶ User interaction
- ▶ Privacy preserving
- ▶ Fairness and biases
- ▶ Interpretability

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Efficiency

- ▶ Develop fast and scalable data mining algorithms (time-efficient)
 - ▶ Effective data structure
 - ▶ Parallel, distributed, and incremental mining
 - ▶ Heuristic and Approximation algorithms
- ▶ Memory-efficient
 - ▶ Develop algorithms that can handle huge amounts of data that cannot be held in RAM.

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Effectiveness

- ▶ Accuracy
 - ▶ How to develop models from data to make accurate predictions on future data
- ▶ Interestingness and actionability of discovered patterns
 - ▶ How to identify interesting patterns from a large number of patterns discovered
 - ▶ How to use the discovered patterns

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User Interaction

- ▶ Interactive mining
 - ▶ Incorporate background knowledge
 - ▶ Combine objective and subjective measures
- ▶ Visualization helps such an integration
 - ▶ Visualization of data
 - ▶ Presentation of mining results
 - ▶ Visualization of the mining process

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Data Mining with Privacy

- ▶ Data may contain private information
 - ▶ Data mining can invade privacy
- ▶ Technical solutions can limit privacy invasion
 - ▶ Replacing sensitive personal data with anon. ID
 - ▶ Alter the data so that real values are obscured
 - ▶ Multi-party computation – distributed data
 - ▶ ...
- ▶ Bayardo & Srikant, Technological Solutions for Protecting Privacy, IEEE Computer, Sep 2003

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Fairness and Bias in AI

- ▶ A model is considered *fair* if errors are distributed similarly across different groups of data
- ▶ Unfairness is usually caused by *bias* in
 - ▶ data, or/and
 - ▶ learning algorithm

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Interpretability

- ▶ How easy to interpret a learned model/pattern
- ▶ How easy for a human to understand the cause of a decision made by a model

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Next Class

- ▶ Frequent pattern and association rule mining
- ▶ Reading: Chapter 4 of Han's book (Pattern mining: basic concepts and methods)

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